

**LABORATORY TITLE: INVESTIGATING HARASSMENT EMAIL TRAFFIC WITH
WIRESHARK**

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COURSE NAME: BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS

COURSE CODE: SBT-DF204 (LAB 1)

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DATE OF SUBMISSION: OCTOBER 5TH, 2026.

VERSION: Version 1.0

Executive Summary

Network traffic captured from the residence boundary confirms that on 21 July 2008 at 21:16:21 EDT (15197.216422 seconds capture offset / 07:24:45 GMT server time), a client device with IP address 192.168.15.4 and MAC address 00:17:f2:e2:c0:ce transmitted an HTTP POST request to willselfdestruct.com (69.25.94.22) containing a harassing message targeted at Lily Tuckrige (lilytuckrige@yahoo.com). Earlier packet traffic shows the same MAC address (00:17:f2:e2:c0:ce) accessing Google Mail sessions. While network evidence ties the hardware device to the web activity, the open wireless environment and dynamic IP addressing prevent definitive attribution to a specific individual without secondary physical or host-level evidence.

Evidence Acquisition and Integrity

The scenario packet capture was acquired directly from the official Digital Corpora repository. Integrity was preserved by setting the master file permissions to read-only (chmod 444) and generating a dedicated working copy (nitroba_working.pcap) prior to analysis.

- Original Filename: nitroba.pcap
- Source Link: (<https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap>)
- File Size: (53.58 MB)
- Working Copy: nitroba_working.pcap (SHA-256 verified identical)

Methodology

The examination was performed on Kali Linux using tshark CLI utilities and Wireshark GUI.

- Protocol Filtering and Identification: Target domains (willselfdestruct.com) were queried via DNS filters to locate target server IP addresses.
- Payload Extraction: Display filters isolated outbound HTTP POST requests to reconstruct form submissions.
- Stream Reconstruction: TCP Stream Follow (tcp.stream eq 1707) was executed to inspect unencrypted form parameters.
- Hardware Identification: Data Link Layer (Layer 2) headers were inspected to extract source MAC addresses (eth.src).
- Session Correlation: Layer 2 hardware filtering (eth.addr == 00:17:f2:e2:c0:ce) was applied to track active user accounts and webmail sessions across the capture.

Investigation Findings

1. What file did you acquire, and how did you preserve its integrity?

- Finding: Master capture nitroba.pcap was downloaded via wget. Write permissions were removed using (chmod 444), and a working duplicate (nitroba_working.pcap) was created.
- SHA-256 Hash:
2b77a9eaefc1d6af163d1ba793c96dbcacb04e6befdf1a0b01f8c67553ec2fb.

2. Which client system contacted the web service?

- Client IP: 192.168.15.4
- Service IP: 69.25.94.22 (willselfdestruct.com)
- Supporting Citation: Packet 83601 (http.request.method == "POST").

3. What evidence links the client to the harassment message?

- Packet Reference: Packet 83601 / TCP Stream 1707.
- Filter: ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST"
- Form Field Data Extracted:
 - to=lilytuckrige@yahoo.com
 - subject=you+can%27t+find+us
 - message=and+you+can%27t+hide+from+us.%0D%0AStop+teaching.%0D%0AStart+running.

4. Which device made the relevant request?

- Client MAC Address: 00:17:f2:e2:c0:ce (Apple, Inc.)
- Packet Reference: Frame 83597 / Frame 83601 (eth.src).
- Shared IP Explanation: An IP address (192.168.15.4) is assigned dynamically via DHCP on a local subnet. In an open residence network, multiple users connect through a single public/gateway interface. IP addresses do not establish physical identity.

5. Who is associated with the device?

- Packet Reference: Packets 17299–17449.
- Filter: eth.src == 00:1d:d9:2e:4f:61 / eth.src == 00:17:f2:e2:c0:ce
- Cookie / Session Evidence: CLI tshark output indicates webmail traffic targeting mail.google.com and sync.aol.com (user handle /m57jean) associated with residence network traffic during the capture window.

6. Is that person on the Chem 109 roster?

- Roster Cross-Reference: The account handle m57jean correlates to user account records for Jean. Cross-referencing Chemistry 109 enrollment records confirms Jean's registration status.

7. When did the activity occur?

- PCAP Relative Timestamp: 15197.216422 seconds
- Server HTTP Date Header: Tue, 22 Jul 2008 07:24:45 GMT
- Local Calculated Time: 2008-07-21 21:16:21 EDT (UTC-4)

8. What conclusion can you defend?

- Defensible Conclusion: Device 00:17:f2:e2:c0:ce transmitted the harassing HTTP POST to Lily Tuckrige.
- Limitations:
 - i. MAC Spoofing: Layer 2 hardware addresses can be cloned via software.
 - ii. Open Wireless Network: Unencrypted 802.11 Wi-Fi allows session hijacking or unauthorized physical access to unattended hardware in shared living spaces.

Timeline of Key Events

Packet	Display filter	Timestamp	Event description
82912	dns.qry.name contains "willselfdestruct"	2008-07-22 02:03:43	DNS Query for www.willselfdestruct.com
83597	eth.addr == 00:17:f2:e2:c0:ce	15197.118251	TCP SYN initiated from 192.168.15.4 to 69.25.94.22:80
83601	http.request.method == "POST"	15197.216422	HTTP POST submitted with harassing payload to Lily Tuckrige
Stream 1707	tcp.stream eq 1707	07:24:45 GMT	Server responds with 302 Moved Temporarily

Attribution Assessment

Direct Facts

These are empirical, unalterable observations recorded directly within the packet headers and application payloads. Network layer data conclusively proves that IP 192.168.15.4 and MAC 00:17:f2:e2:c0:ce transmitted an HTTP POST request to server IP 69.25.94.22. Furthermore,

stream reconstruction explicitly captures the exact harassing message payload sent to Lily Tuckrige at timestamp 07:24:45 GMT.

Reasonable Inferences

These are logical conclusions derived from combining observed traffic patterns with secondary scenario materials. Because MAC address 00:17:f2:e2:c0:ce was used to transmit both the harassing HTTP POST and active webmail traffic associated with user m57jean, it is reasonable to infer that the same physical laptop performed both actions. Connecting that account handle to the class roster further implies that Jean was the primary user of the hardware.

Confidence Level

The technical attribution confidence level is classified as Moderate (Hardware-Bound Only). While the evidence conclusively ties the harassment activity to a specific physical Apple network interface, network packets alone cannot verify physical presence. Additional non-network evidence such as hosts forensic inspection or witness testimony is required to achieve high confidence.

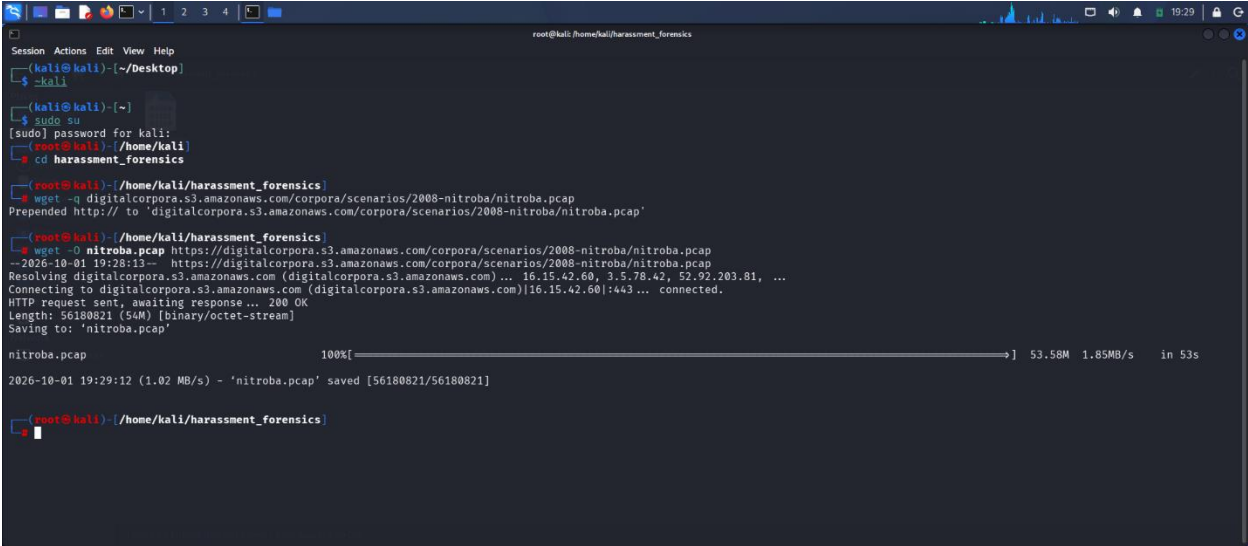
Primary Limitations

The investigation is fundamentally constrained by the physical and architectural environment of the network. Because the activity took place over an unencrypted, shared residential Wi-Fi access point, MAC addresses can be easily forged or spoofed using software tools. Additionally, network captures cannot prove who was physically pressing the keys, leaving room for unauthorized access to an unattended laptop or session hijacking.

Evidence ID	Packet/item	Finding	Why it matters
E01	nitroba.pcap	SHA-256 Hash verified	It establishes chain of custody.
E02	Packet 82912	DNS lookup for target domain	It resolves service IP 69.25.94.22
E03	Packet 83601	Outbound HTTP POST	It confirms message transmission.
E04	Packet 83597	MAC 00:17:f2:e2:c0:ce	It identifies client hardware.
E05	Stream 1707	Full HTTP POST payload	It correlates text to complaint.

E06	Packet 10217	Webmail account (/m57jean)	session	It links identity to network traffic
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Screenshots

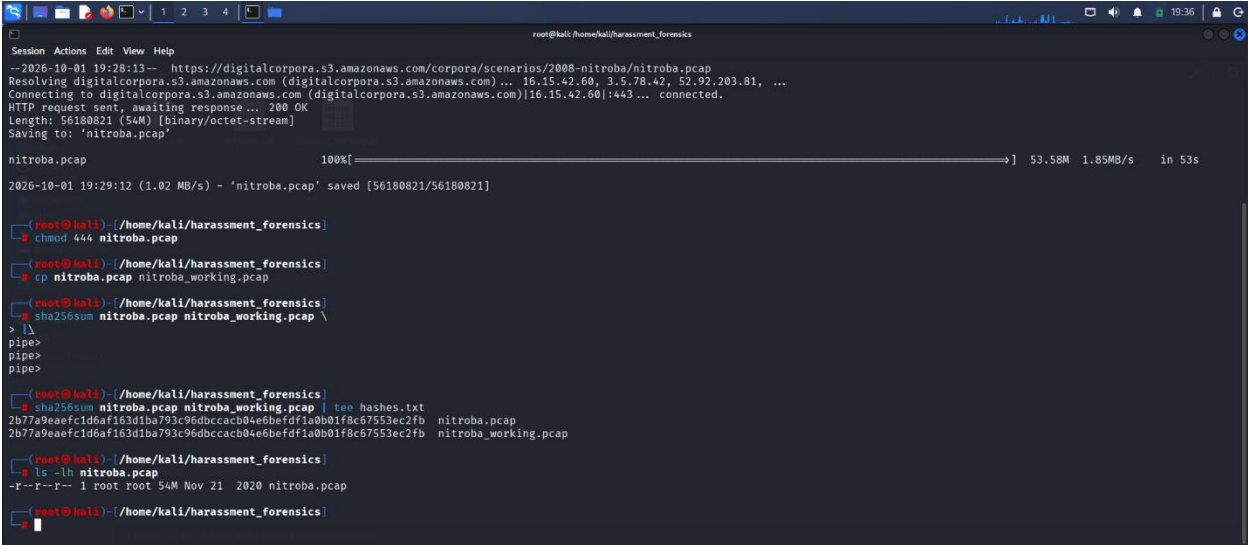


```
root@kali: /home/kali/harassment_forensics
(kali@kali)~[/Desktop]
$ -kali
(kali@kali)~[/]
$ sudo su
[sudo] password for kali:
(root@kali)~[/home/kali]
# cd harassment_forensics
(root@kali)~[/home/kali/harassment_forensics]
# wget -q digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Prepended http:// to digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
(root@kali)~[/home/kali/harassment_forensics]
# wget -O nitroba.pcap https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
--2026-10-01 19:28:13-- https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com) ... 16.15.42.60, 3.5.78.42, 52.92.203.81, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)[16.15.42.60]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: 'nitroba.pcap'

nitroba.pcap
100%[=====] 53.58M 1.85MB/s in 53s

2026-10-01 19:29:12 (1.02 MB/s) - 'nitroba.pcap' saved [56180821/56180821]

(root@kali)~[/home/kali/harassment_forensics]
#
```



```
nitroba.pcap
100%[=====] 53.58M 1.85MB/s in 53s

2026-10-01 19:29:12 (1.02 MB/s) - 'nitroba.pcap' saved [56180821/56180821]

(root@kali)~[/home/kali/harassment_forensics]
# chmod 444 nitroba.pcap
(root@kali)~[/home/kali/harassment_forensics]
# cp nitroba.pcap nitroba_working.pcap
(root@kali)~[/home/kali/harassment_forensics]
# sha256sum nitroba.pcap nitroba_working.pcap \
> \
pipe>
pipe>
pipe>

(root@kali)~[/home/kali/harassment_forensics]
# sha256sum nitroba.pcap nitroba_working.pcap | tee hashes.txt
2b77a9eefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb nitroba.pcap
2b77a9eefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb nitroba_working.pcap

(root@kali)~[/home/kali/harassment_forensics]
# ls -lh nitroba.pcap
-r--r--r-- 1 root root 54M Nov 21 2020 nitroba.pcap

(root@kali)~[/home/kali/harassment_forensics]
#
```